VIDEO GAME SALES DASHBOARD

A Data Analytics Project Report

Anudip Foundation

In Partial Fulfillment of the Requirements for the Completion of
Data Analytics / Power BI Training Program

Batch Code: 4846

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Date of Submission	Place
20.03.2025	Chennai

1. ABSTRACT

In the modern data-driven world, analyzing large datasets is essential for identifying trends and making informed decisions. This project focuses on analyzing global video game sales data using Power BI. The objective is to transform raw data into meaningful insights through interactive visualizations.

The dashboard highlights key aspects such as genre-wise sales, platform performance, regional distribution, and yearly trends. By analyzing these factors, the project provides a comprehensive understanding of the gaming industry and its market behavior.

2. INTRODUCTION

2.1 Background

The video game industry is one of the fastest-growing entertainment sectors globally. With millions of games released across multiple platforms, analyzing sales data helps in understanding consumer preferences and market trends.

2.2 Purpose of the Project

This project aims to:

- Convert raw sales data into meaningful insights
- Identify key trends in the gaming industry
- Provide a visual representation using Power BI

2.3 Scope

The project focuses on:

- Historical sales data analysis
- Genre and platform comparison
- Regional market insights
- Dashboard-based visualization

3. OBJECTIVES

The main objectives of this project are:

- To analyze global video game sales data
- To identify top-performing genres and platforms
- To study regional sales distribution
- To visualize trends using Power BI
- To build an interactive dashboard for better understanding

4. LITERATURE REVIEW

Data visualization tools like Power BI have become essential in business analytics. Previous studies show that dashboards improve decision-making by presenting data in a clear and interactive format.

Gaming industry analysis reports indicate that:

- Action games dominate global sales
- Console platforms contribute significantly to revenue
- Regional preferences vary across markets

This project builds upon these findings using real dataset analysis.

5. DATASET DESCRIPTION

The dataset used in this project contains historical video game sales data.

5.1 Data Attributes

Column Name	Description
Name	Game title
Platform	Gaming platform (PS, Xbox, etc.)
Year	Release year

Column Name	Description
Genre	Type of game
Publisher	Company that published the game
NA Sales	Sales in North America
EU Sales	Sales in Europe
JP Sales	Sales in Japan
Global Sales	Total worldwide sales

5.2 Data Characteristics

- Time period: 1997 – 2016
- Data type: Structured (CSV format)
- Contains multiple categories and regions

6. TOOLS AND TECHNOLOGIES USED

6.1 Power BI

Power BI is a business analytics tool used to create interactive dashboards and reports.

6.2 CSV Dataset

The dataset is stored in CSV format, making it easy to import and process.

6.3 Data Processing

- Data cleaning
- Data transformation
- Visualization

7. METHODOLOGY

The project follows a systematic approach:

Step 1: Data Collection

The dataset is collected in CSV format containing video game sales information.

Step 2: Data Cleaning

- Removed missing or inconsistent values
- Ensured correct data types

Step 3: Data Import

- Imported dataset into Power BI

Step 4: Data Transformation

- Created calculated fields
- Aggregated data for analysis

Step 5: Visualization

- Created charts (bar, line, pie, etc.)
- Designed dashboard layout

Step 6: Analysis

- Interpreted patterns and trends
- Derived meaningful insights

8. DASHBOARD DESIGN

The dashboard consists of multiple components:

8.1 KPI Cards

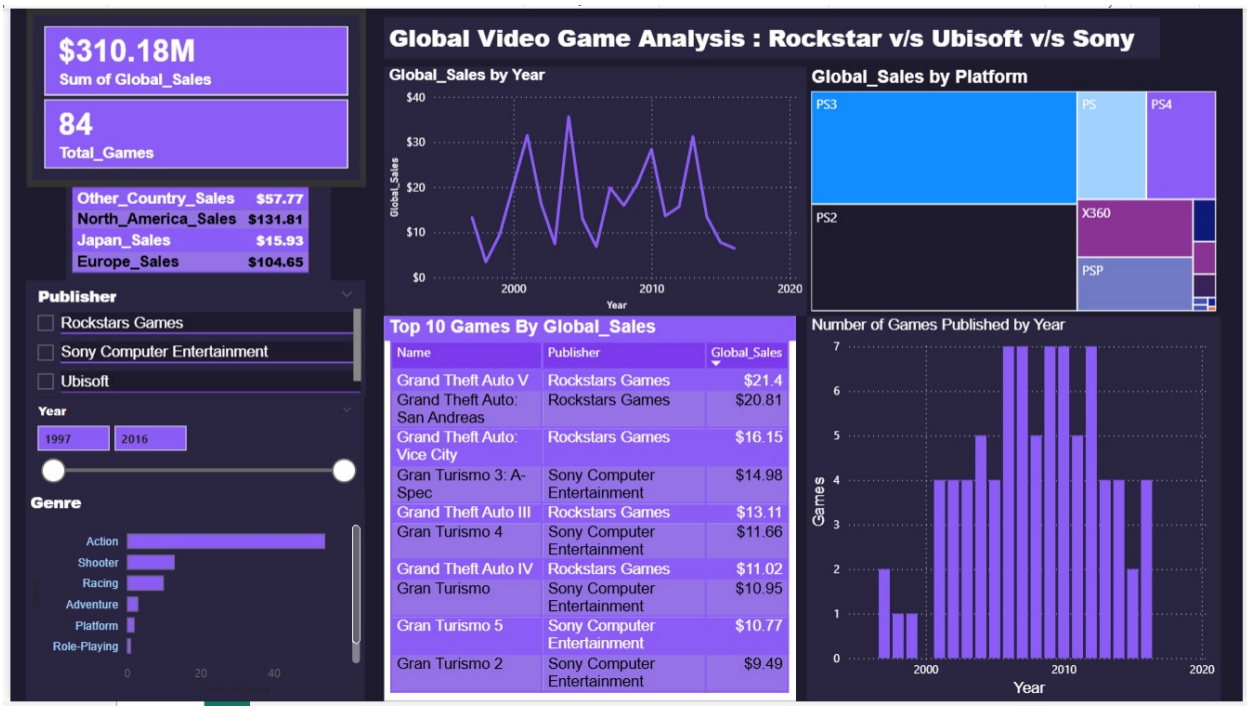
- Total Global Sales
- Average Sales
- Total Number of Games

8.2 Charts Used

- Bar Chart → Genre and platform comparison
- Line Chart → Year-wise sales trend
- Pie Chart → Regional distribution
- Table → Publisher analysis

8.3 Layout

- Top: Key performance indicators
- Middle: Trend analysis
- Bottom: Detailed comparisons



9. DATA ANALYSIS & INTERPRETATION

9.1 Genre Analysis

- Action genre shows highest sales
- Other genres contribute less significantly

9.2 Platform Analysis

- PlayStation platforms dominate

- Older platforms still contribute to sales

9.3 Year-wise Trends

- Peak sales observed between 2005–2012
- Slight decline afterward

9.4 Regional Analysis

- North America leads in sales
- Europe follows closely
- Japan contributes less compared to others

9.5 Publisher Analysis

- Few publishers dominate the market
- Indicates high market concentration

10. KEY INSIGHTS

- The gaming industry is **hit-driven**
- Action games dominate global sales
- PlayStation platforms lead market share
- Western regions contribute most revenue
- Sales peaked in early 2010s

11. ADVANTAGES OF THE SYSTEM

- Easy visualization of complex data
- Interactive filtering and analysis
- Helps in decision-making
- Provides clear insights into trends

12. LIMITATIONS

- Dataset is limited to historical data
- No real-time updates
- Limited regional breakdown

13. FUTURE SCOPE

- Integration with real-time data sources
- Use of machine learning for predictions
- Expansion to include more recent data
- Enhanced dashboard interactivity

14. CONCLUSION

The project successfully demonstrates how Power BI can be used to analyze and visualize video game sales data. The dashboard provides valuable insights into industry trends, helping users understand market behavior effectively.

Overall, this project highlights the importance of data visualization in transforming raw data into meaningful information.

15. REFERENCES

- Kaggle Dataset (Video Game Sales)
- Power BI Documentation
- Online tutorials and learning resources